

NEB Release 1: Note

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- Merit 1. no hessian is needed. By using only energies and gradients, you can get a structure close to the transition-state structure.
- Merit 2. not necessary to guess the transition-state structure. You need to give the structures of the reactant and product.
- Merit 3. automatically avoid overlapping atoms along an initial path guessed by the structures of the reactant and product. There is no problem for Mark's example, but as you may understand, some atoms are often overlapped along the linear synchronous transit.
- Merit 4. no angular momentum generated by the geometrical displacement between the reactant and product. This is sometimes important. As you understand, no angular momentum is generated along exact IRCs. The present NEB path has only 10 or more points, so the angular momentum should be erased to get an appropriate path.
- Demerit 1. it is difficult to define the vector along the reaction path. So, as you know, NEB paths are generally different from IRCs.
- Demerit 2. very slow to converge. This is caused by the demerit 1.
→ some optimizers will be examined. DIIS & DIIS+BFGS are known to be ineffective so far.
- Demerit 3. no exact transition-state structure. If you can have a hessian, then, you can submit RUNTYP=sadpoint by using the structure at the energetically highest bead or the structure given by the structures of two highest beads.

NEB Release 2 can give you a structure very close to the true transition-state structure without hessian. We already have the code, but it is independent or isolated. So, we need to work on adding it to GAMESS. The problem is that Dmitri has his own code similar to ours and his code can do FMO.

```
C> @brief    NEBPATH main driver, Release 1
C>
C> @details  Nudged Elastic Band (NEB) method (Release 1, April 2019)
C>           find a reaction path for specific reactant and product.
C>           Multiple transition states can be located along an NEB path.
C>
C>           PLAN: Release 2 --- Climbing NEB method to locate a transition state.
C>                        (July 2019)
C>           PLAN: Release 3 --- String method to locate a transition state exactly.
C>                        (December 2019)
C>
C> @author    Nozomi Sawada, Mamoru Haruta, Toshio Asada, Shiro Koseki
C>           Osaka Prefecture University, Japan
```

Manual

```
c *****
C Options for $NEB group
c *****
C Caution: The coordinates of the reactant are given by $DATA
c           The coordinates of the product should be given by $DATAPD.
c           The atom order should be same in both reactant and product.
c           The product should be aligned as close to the reactant as possible,
c           Otherwise, some curious paths could be generated by this routine.
c           The molecular orientation is very important to obtain a reasonable
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c      reaction path.
c      *****
c
c      Basically, molecular symmetry cannot be used.  However, if a reaction
c      path has symmetry on the whole way, you can set MOLSYM=.true.
c
c      MAXCYC is set to 500 as a default, but, generally speaking, the convergence
c      is currently very slow for the optimization of perpendicular gradient.  So,
c      You may need to restart your job several times.  For restart, $NEB and
c      $BEADS groups in *.dat file of the previous run need to be pasted into your
c      input file.
c
c      It would be difficult to reach the condition that all energy gradients are smaller
c      than opttol.  You may decide in searching a saddle point, after the plot of
c      gradients along the path has two peaks (the minimum between two peaks could
c      be the structure close to a target transition-state structure).  If you could not
c      use hessian, you have to wait for the release 2 or 3 of this source code.
c
c      Caution: keywords can have only 6 characters.
c
c      $NEB  NBEADS      = number of beads along the NEB path (default=15)
c              (maximum number of beads is set to MXBEADS=50)
c      untcrd  = unit for the coordinates given by the $DATAPD and $DATAVI
c              groups; angstroms or bohr (default = angstroms)
c
c      MAXCYC      = maximum number of cycles for geometry optimization
c              along the path perpendicular to the reaction path.
c              (default = 500)
c
c      OPTTOL      = Tolerance for the gradient RMS perpendicular to the
c              reaction path (unit=au/bohr, default = 0.001d+00)
c              (0.0001d+00 is too small?)
c
c      QUICKDT     = distance displaced from each bead along the path
c              perpendicular to the reaction path.
c              = (default = 0.2d+00) (unit = bohr)
c
c      SPFORC      = force constant for a spring between two adjacent beads
c              (unit = hartree/bohr**2, default = 0.5d+00)
c
c      VIAPNT      = a via-point structure along the reaction path
c              The coordinates are given in the $DATAVI group.
c              (default=.false.)
c
c      RESTRT      = Restart option.  See $BEADSX.
c              (default = .false.)
c
c      iSTART      = Number of cycles in the previous calculation
c              (default = 0)
c
c      ----- These options should not be listed in the manual...
c      Morint      = caution for molecular orientation
c              (default=.false.)
c      MOLSYM      = Say .true. if your reaction path has molecular symmetry
c              along the whole way.  Otherwise, only "C1" can be used
c              in the $DATA group.
c              (default=.false.)
c      -----
c
c      The followings are unavailable right now
c
c      TSLOCT      = locating a transition state structure
c              (default = .false.)

```

```

c
c      OPTMZR   = method of a geometrical structure at each bead
c                quick-min   (default)
c                DIISBFGS
c                steepest descent (not ready)
c                FIRE (not ready)
c
c      $DATAPD   = the coordinates of the product
c                  atom name, atomic number, x, y, z for each atom
c                  (should be given for all atoms, even if a molecular symmetry exists)
c
c      $DATAVI   = the coordinates of the via-point structure (optional)
c                  atom name, atomic number, x, y, z for each atom
c                  (should be given for all atoms, even if a molecular symmetry exists)

```

Description of parameters –

#Bead = number of beads

S_{total} = distance from the reactant $\left[\sqrt{\text{amu}} \cdot \text{bohr} \right]$

dS = distance from the previous bead $\left[\sqrt{\text{amu}} \cdot \text{bohr} \right]$

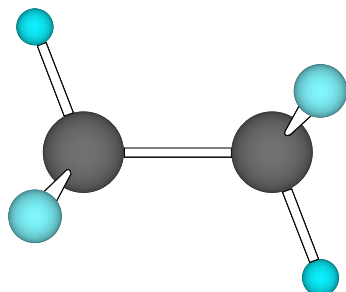
dE = Energy relative to the reactant [kcal/mol]

GRMS = root-mean-square of the gradient [kcal/mol/angs]

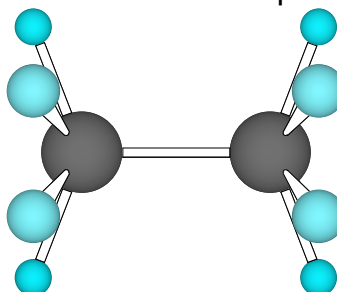
P. GRMS = root-mean-square of the gradient perpendicular to the reaction path [kcal/mol/angs]

Exam01. C–C rotation in ethane

Given orientation of reactant and product



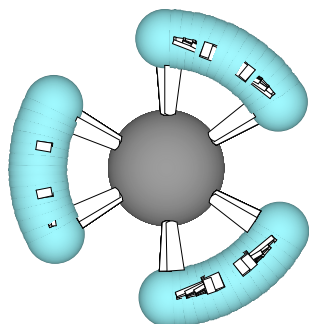
Final orientation before optimization



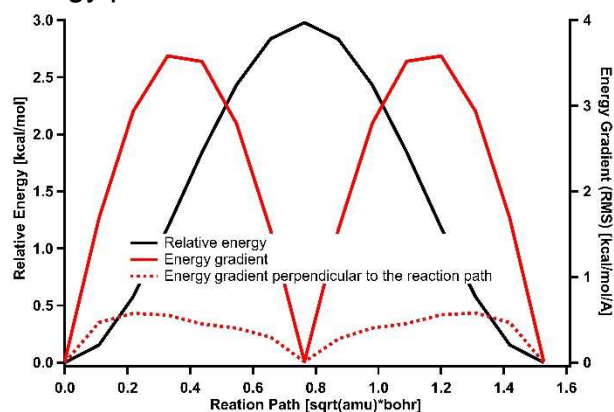
#cycles = 535

#Bead	Dist & dD [$\sqrt{\text{amu}} \cdot \text{bohr}$]		dE (kcal/mol)	G norm (kcal/mol/Å)	P. gradient(kcal/mol/Å)
1	0.000000	0.000000	0.000	0.019076	0.000000
2	0.108630	0.108630	0.153	1.684231	0.468874
3	0.217940	0.109310	0.577	2.935729	0.574878
4	0.327786	0.109845	1.182	3.581841	0.553807
5	0.437722	0.109937	1.844	3.516669	0.450054
6	0.547395	0.109673	2.433	2.790905	0.402882
7	0.656306	0.108911	2.836	1.545611	0.296697
8	0.764097	0.107790	2.978	0.010450	0.009702
9	0.871885	0.107788	2.834	1.548417	0.275426
10	0.980722	0.108837	2.431	2.794894	0.403558
11	1.090308	0.109586	1.842	3.518710	0.455979
12	1.200154	0.109846	1.181	3.581574	0.556856
13	1.309902	0.109749	0.576	2.935249	0.579811
14	1.419132	0.109230	0.153	1.682347	0.465847
15	1.527682	0.108550	-0.000	0.009568	0.000000

Reaction path



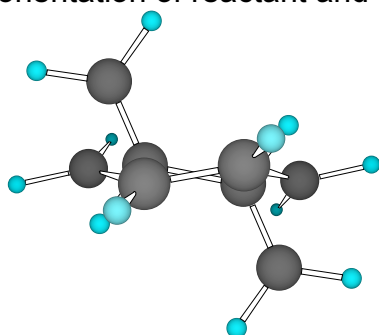
Energy profile



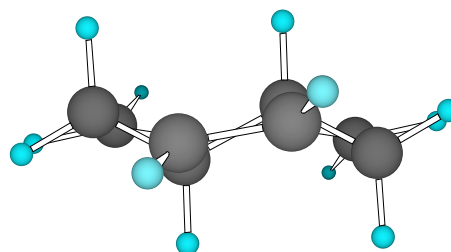
→ runtyp=sadpoint gives -79.234115 hartree with an imaginary frequency of $i315\text{ cm}^{-1}$

Exam02. cis-trans isomerization in butadiene

Given orientation of reactant and product

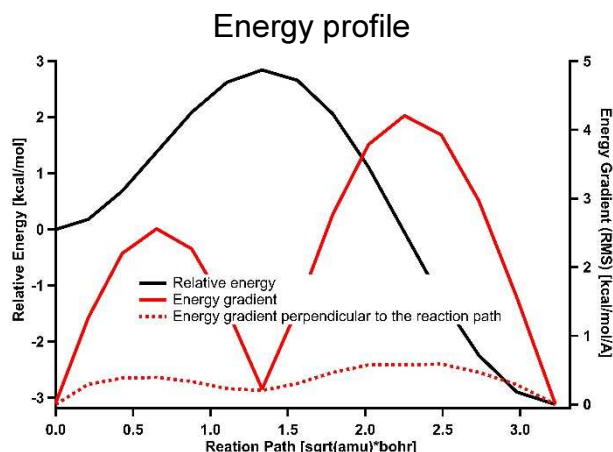
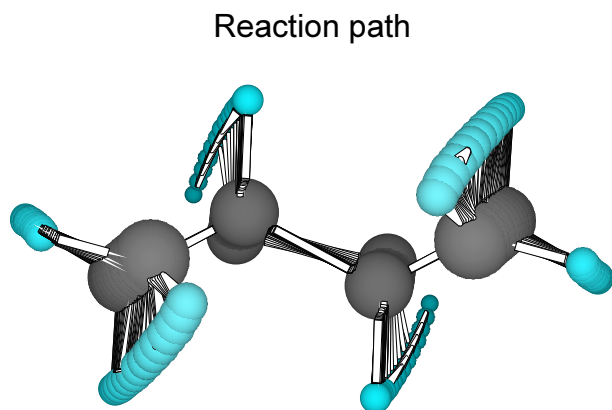


Final orientation before optimization



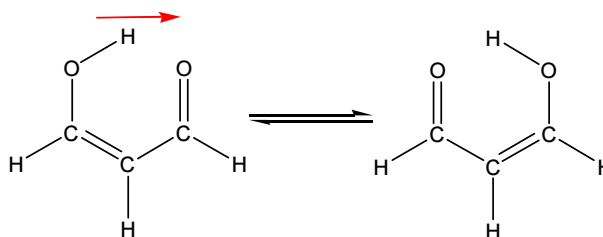
#cycles = 1096

#Bead	Dist & dD [sqrt(amu)*bohr]		dE (kcal/mol)	G norm (kcal/mol/Å)	P. grad(kcal/mol/Å)
1	0.000000	0.000000	0.000	0.020462	0.000000
2	0.210435	0.210435	0.180	1.262463	0.292481
3	0.428689	0.218253	0.684	2.198161	0.383463
4	0.652188	0.223500	1.380	2.557778	0.393856
5	0.878515	0.226326	2.087	2.266737	0.335303
6	1.105799	0.227285	2.624	1.391061	0.230695
7	1.333470	0.227671	2.843	0.223560	0.203785
8	1.561074	0.227603	2.660	1.395288	0.305074
9	1.790101	0.229028	2.061	2.766787	0.466403
10	2.020819	0.230718	1.112	3.785586	0.576018
11	2.253470	0.232650	-0.053	4.206345	0.575985
12	2.491057	0.237588	-1.238	3.924673	0.588199
13	2.732759	0.241701	-2.240	2.970487	0.468271
14	2.978521	0.245762	-2.896	1.563468	0.288327
15	3.228206	0.249685	-3.119	0.004686	0.000000



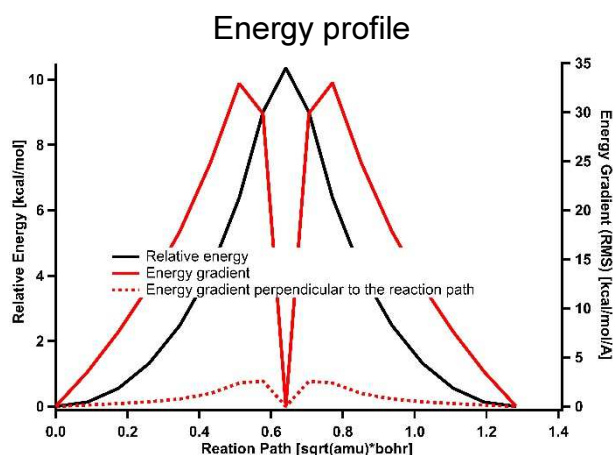
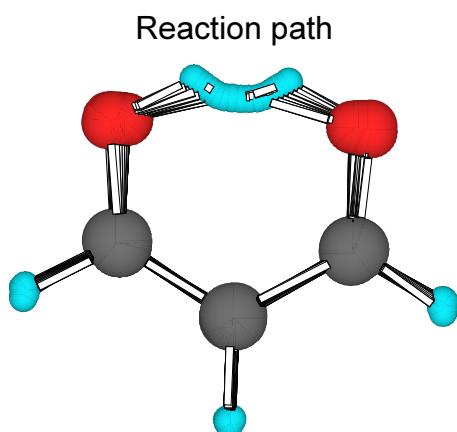
→ runtyp=sadpoint gives -154.916223 hartree with an imaginary frequency of $i175\text{ cm}^{-1}$

Exam03. 1,5-H migration in malon-aldehyde



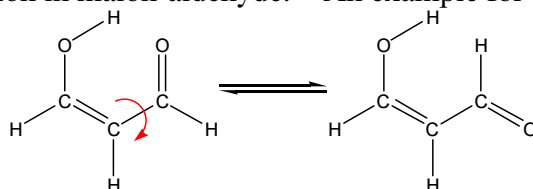
#cycles = 900

#Bead	Dist & dD [sqrt(amu)*bohr]	dE (kcal/mol)	G RMS (kcal/mol/A)	P. GRMS(kcal/mol/A)
1	0.000000 0.000000	0.000	0.002005	0.000000
2	0.088569 0.088569	0.132	3.528188	0.137327
3	0.174864 0.086295	0.555	7.611419	0.291096
4	0.260360 0.085495	1.320	12.310819	0.475797
5	0.345634 0.085274	2.478	17.838396	0.756252
6	0.431014 0.085380	4.132	24.849815	1.322575
7	0.511016 0.080002	6.378	32.942751	2.382186
8	0.577340 0.066324	9.007	29.847421	2.583852
9	0.641236 0.063896	10.352	0.005203	0.000919
10	0.705415 0.064179	9.000	29.884754	2.573424
11	0.771028 0.065613	6.390	32.985707	2.381682
12	0.851569 0.080542	4.124	24.810321	1.323005
13	0.936380 0.084811	2.483	17.861543	0.756079
14	1.022123 0.085743	1.318	12.299247	0.476206
15	1.107282 0.085159	0.556	7.619345	0.290881
16	1.193777 0.086495	0.131	3.526157	0.137519
17	1.282257 0.088480	0.000	0.022145	0.000000



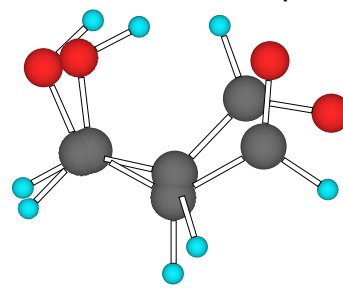
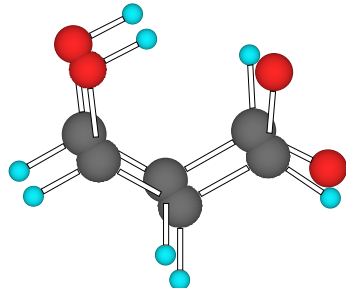
→ runtyp=sadpoint gives -265.636478 hartree with an imaginary frequency of $i1787\text{ cm}^{-1}$

Exam04. cis-trans isomerization in malon-aldehyde. An example for viapnt=.true



Given orientation of reactant and product

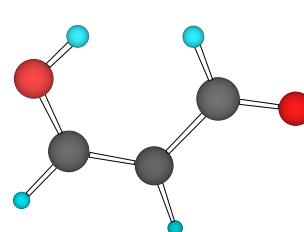
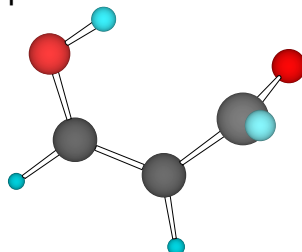
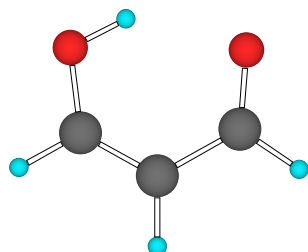
Final orientation before optimization



Reactant

via-point

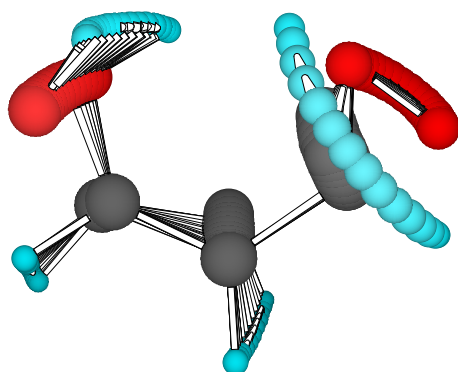
Product



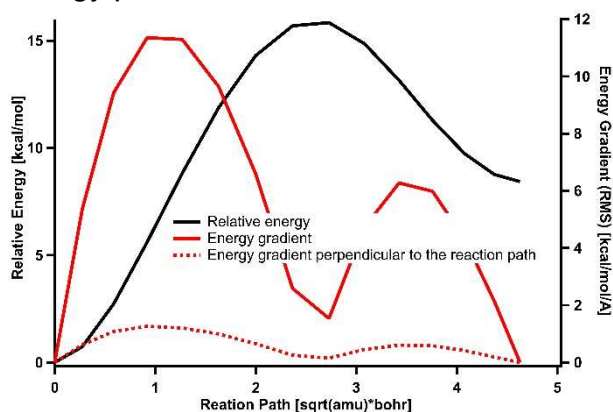
#cycles = 5000

#Bead	Dist & dD [sqrt(amu)*bohr]		dE (kcal/mol)	G RMS (kcal/mol/A)		P. GRMS(kcal/mol/A)
1	0.000000	0.000000	0.000	0.002005		0.000000
2	0.273365	0.273365	0.730	5.357300		0.616043
3	0.586791	0.313426	2.732	9.429540		1.082027
4	0.919077	0.332286	5.609	11.352494		1.262365
5	1.269441	0.350364	8.823	11.289147		1.205493
6	1.631044	0.361603	11.877	9.641515		0.994434
7	1.998955	0.367910	14.302	6.593541		0.654248
8	2.366648	0.367693	15.693	2.585827		0.251043
9	2.729022	0.362374	15.840	1.538763		0.147407
10	3.082200	0.353178	14.856	4.732913		0.453318
11	3.426180	0.343981	13.153	6.271982		0.605506
12	3.759496	0.333316	11.280	5.981792		0.584211
13	4.078777	0.319281	9.731	4.314441		0.414184
14	4.373738	0.294961	8.760	2.140918		0.187591
15	4.628406	0.254668	8.431	0.002994		0.000000

Reaction path



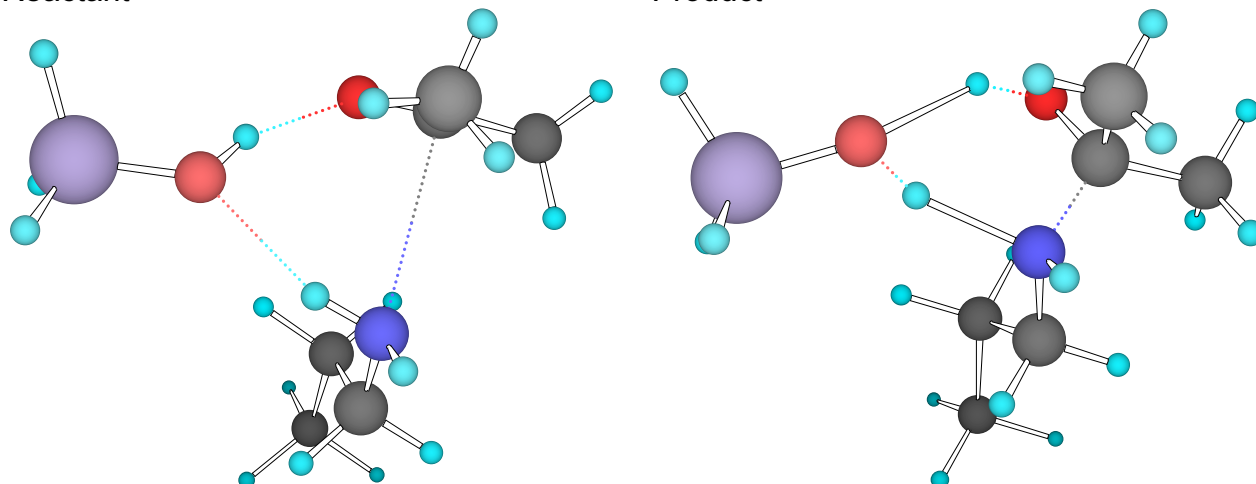
Energy profile



→ exam04r is an example for the input of the restarting job.

→ runtyp=sadpoint gives -265.627588 hartree with an imaginary frequency of $i193\text{ cm}^{-1}$

Exam05. Mark's sample input (MP2) (DFT). See J. Phys. Chem. B 2016, 120, 1660–1669.
 Reactant Product



•Summary (MP2/6-31+G(d,p))

#cycles = 116

#Bead	S_{total}	dS	dE	GRMS	P. GRMS
1	0.000000	0.000000	0.000	0.035510	0.000000
2	0.205667	0.205667	0.126	0.119803	0.095941
3	0.412299	0.206632	0.538	0.258141	0.209166
4	0.619395	0.207095	1.241	0.407920	0.332149
5	0.829726	0.210332	2.327	0.608531	0.495509
6	1.044809	0.215083	3.958	0.873557	0.573155
7	1.283517	0.238708	7.608	1.370832	0.586422
8	1.518780	0.235263	12.441	1.278002	0.698018
9	1.733316	0.214536	13.187	1.601472	0.869767
10	1.936736	0.203419	7.551	1.556512	0.882705
11	2.132742	0.196006	3.060	1.619787	0.663342
12	2.318375	0.185633	-2.235	1.090639	0.555274
13	2.524917	0.206541	-4.819	0.516675	0.317214
14	2.730008	0.205091	-5.747	0.287447	0.231922
15	2.935954	0.205946	-6.077	0.028778	0.000000

→ runtyp=sadpoint was failed. It is because GRMS does not give you two mountains. This means that Bead #9 is not close to the true transition yet. You have to continue the NEB calculations in order to obtain two mountains for GRMS.

•Summary (DFT(M06-2X)/6-31+G(d,p))

#Bead	S_{total}	dS	dE	GRMS	P. GRMS
1	0.000000	0.000000	0.000	0.038274	0.000000
2	0.171707	0.171707	0.151	1.445988	0.121263
3	0.343878	0.172171	0.590	3.017714	0.259803
4	0.515667	0.171789	1.371	4.992871	0.421128
5	0.693847	0.178180	2.731	7.878600	0.624847
6	0.886697	0.192850	5.068	11.482215	0.725958
7	1.089681	0.202984	8.695	12.137290	0.673687
8	1.287218	0.197537	11.668	9.376075	0.761707
9	1.443212	0.155995	11.094	18.032578	1.184814
10	1.624868	0.181656	6.272	16.166691	1.012060
11	1.770437	0.145568	0.952	22.950117	0.945348
12	1.936346	0.165910	-4.793	10.295849	0.512494
13	2.115322	0.178976	-6.822	5.501602	0.402053
14	2.284535	0.169213	-7.839	2.928355	0.226041

15 2.453757 0.169222 -8.192 0.028085 0.000000

→ runtyp=sadpoint gives -734.5640581 hartree with an imaginary frequency of $i139\text{ cm}^{-1}$
 (many steps for the saddle-point search were needed, because GRMS at Bead #8 is not small enough to search a TS yet?)

